

MA 578 Final Project: Diabetes Pedigree Modeling

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1 Introduction

Diabetes is a globally prominent health condition with no definitive cure, yet type 2 diabetes is largely preventable and can be delayed through proper treatment. It is chiefly important in the field of healthcare to understand early warning signs for a patient's risk of diabetes through biometric data and patient history. The objective of this study is to predict based on diagnostic measurements a patient's risk of diabetes based on available data. In addition, we would like to understand the particular relationship between patient data and their resulting risk. Both of these goals can be met through the design and analysis of a Bayesian regression model. For the purpose of training a model, we will use a dataset from the National Institute of Diabetes and Digestive and Kidney Diseases. The dataset contains diagnostic measurements from 768 women aged 21 or older of Pima Indian descent, including blood pressure, glucose levels, insulin levels, and their diabetes pedigree function (DPF), which quantifies ones imposed health risk due to diabetes based on family history. We wish to see if DPF can be predicted using correlative factors if a patient's family history is not available.

Index	Pregnancies	Glucose	Blood Pressure	BMI	Age	Pedigree
1	6	148	72	33.6	50	0.627
2	1	85	66	26.6	31	0.351
3	8	183	64	23.3	32	0.672
...	...	...	...	...	...	...

Table 1: Dataset Head

Although the dataset may provide insight applicable to the population at large, conclusions from this study will only regard the population of Pima Indian women aged 21 and older due to the sampling focus. The dataset has been cleaned with the exception of missing values. Because the portion of patients in the dataset with missing values is very small, they will be removed to simplify analysis without the risk of discarding a large amount of information.

2 Analysis

For the purpose of performing prediction analysis, the dataset was first split into training and validation sets. For the validation set, 100 patients were chosen uniformly at random and the remaining patients were put into the training set that will be used to fit a Bayesian linear regression model. Accordingly, the sampling model for the pedigree outcome, y , will be given by a regression model with Gaussian noise that considers all available predictors: number of pregnancies (x_1), plasma glucose levels from an oral glucose tolerance test (x_2), blood pressure in mm Hg (x_3),

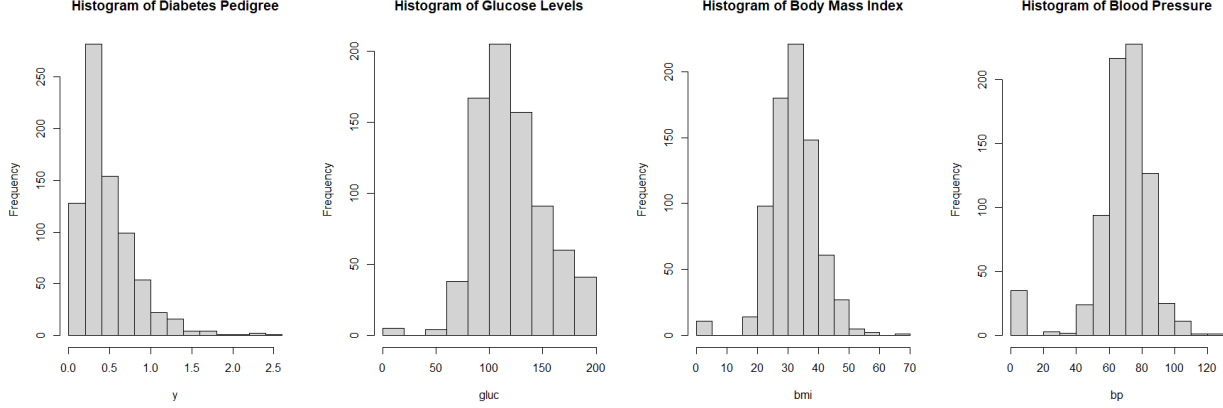


Figure 1: Variable Distributions

triceps skin fold thickness in mm (x_4), 2-hour serum insulin levels in $\mu\text{U}/\text{mm}$ (x_5), body mass index (x_6) and age in years (x_7).

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, i = 1, \dots, n$$

$$y|X, \beta \sim \mathcal{N}(X\beta, \sigma^2 I)$$

It is unlikely that every predictor available contributes meaningfully to the overall model, so we will impose sparsity by using a spike and slab prior for each coefficient (excluding the intercept). This will introduce an inclusion parameter z , conditional on which we will utilize Zellner's g-prior.

$$\beta_j = z_j b_j, j = 1, \dots, p$$

$$z_j \overset{iid}{\sim} \text{Bern}\left(\frac{1}{2}\right)$$

$$\sigma^2 \sim \text{IG}\left(\frac{\nu_0}{2}, \frac{\nu_0}{2} \sigma_0^2\right)$$

$$\beta|X, z, \sigma^2 \sim \mathcal{N}(0, g\sigma^2[X_z^\top X_z]^{-1})$$

Due to a lack of external information, we will initially specify a relatively uninformative prior with $\nu_0 = 1$, $g = n$, $\sigma_0^2 = 5$ so that the prior can be interpreted as coming from a single unit sample. The choice of sampling model and prior will yield the following marginal and full conditional distributions:

$$\sigma^2|y, X, z \sim \text{IG}\left(\frac{\nu_0+n}{2}, \frac{\nu_0\sigma_0^2+SSR_g^z}{2}\right)$$

$$\beta_z|\sigma^2, y, X, z \sim \mathcal{N}\left(\frac{g}{g+1}\hat{b}_z, \frac{g}{g+1}\sigma^2[X_z^\top X_z]^{-1}\right)$$

$$p(z_j = 1|X, y, z_{-j}) = \frac{o_j}{1+o_j} \propto p(y|X, z) * p(z)$$

which will allow us to implement a Gibbs sampler that draws from the desired posterior $p(\beta, \sigma^2|y, X)$ (see Algorithm 1). Our goal is to construct a Bayes-optimal predictive model for the diabetes pedigree of a patient, which we will achieve through Bayesian Model Averaging (BMA). In addition, we would like a clean interpretable model, so we will also construct a Median Probability Model (MPM) using estimated Posterior Inclusion Probabilities (PIP) from the Gibbs sampler. We ran the Gibbs sampler for 1,000 iterations, which yields an effective sample size of at least 400 and a reasonably stable Monte Carlo standard error that provides between 3 and 5 digits of accuracy (see Table 2).

We may reasonably assume that the Gibbs sampler is adequate as the sample ACF for each parameter decays exponentially and is insignificant after at most 5 lags (see Figure 2). In addition, there are no visible trends in the trace plot for σ^2 , or for the coefficient parameters with higher

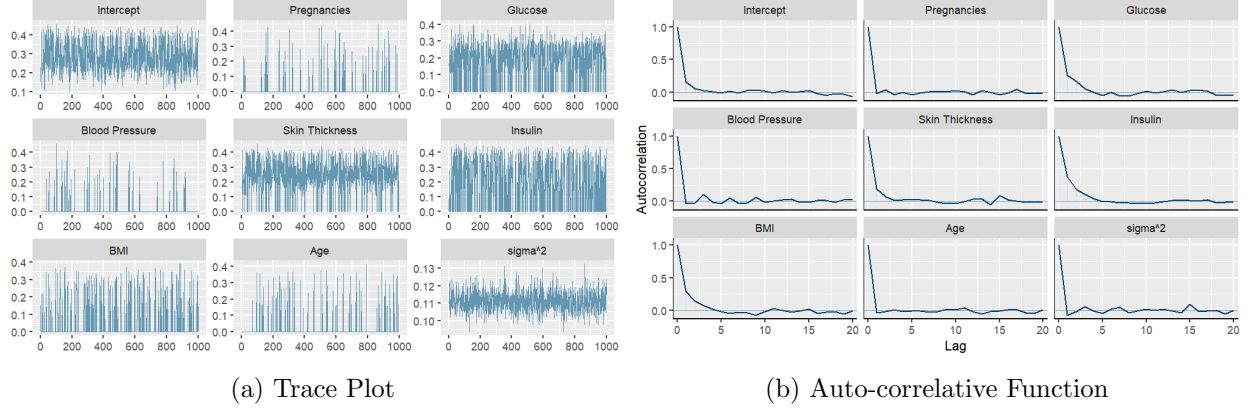


Figure 2: MCMC Diagnostic Plots

posterior inclusion probabilities. The other coefficients are frequently evaluated to be zero, around which the chains feature prominent "stickiness," however, this is a result of the z parameter which imposes sparsity. Using the sampled values of β, σ^2 , replicate datasets were generated to perform posterior predictive checking on the pedigree outcome distribution (see Figure 3(a)).

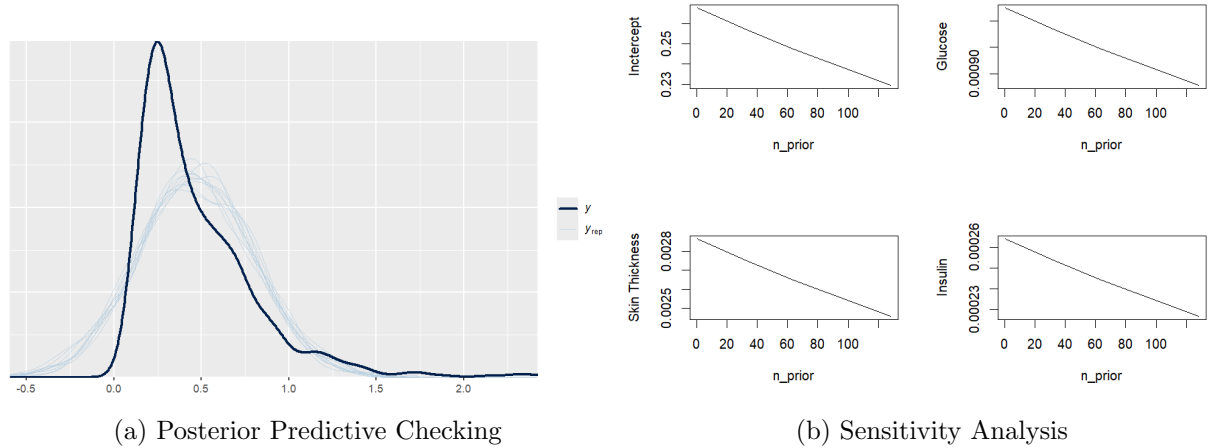


Figure 3: Model Checking

Notably, the observed distribution in y deviates from the replicated y^{rep} distributions due to a rightward skew and higher kurtosis. This suggests that the distribution $p(y|\beta, X)$ may not be Gaussian as the sampling model assumes. However, the sampling model will be kept for two reasons. First, the deviation is not significant enough to warrant a different sampling model that would complicate design and analysis. Second, even if the posterior predictive model is not adequate, the Gibbs sampler still provides a good estimate of the marginal distribution $p(z|X, y)$, which can be utilized to fit a separate more adequate model with the MPM predictors.

In addition to posterior predictive checking, sensitivity analysis was performed by computing the posterior expectation of each β coefficient included in the MPM model based on a range of prior sample size values from 1 to 128 ($n_{prior} = \frac{n}{g}$). As expected, a larger prior sample size imposes linear shrinkage on the posterior expectation (see Figure 3(b)). In this case, a small prior sample size which does not impose shrinkage seems adequate because the spike and slab nature of the prior

Parameter	$\hat{\beta}_{BMA}$	$\hat{\beta}_{MPM}$	MCSE	PIP	n_{eff}
β_0 (Intercept)	0.2831	0.2676	0.0029	1.0000	729.47
β_1 (Pregnancies)	0.0134	.	0.0020	0.0460	1000.0
β_2 (Glucose)	0.1571	0.0102	0.0057	0.0057	476.13
β_3 (Blood Pressure)	0.0110	.	0.0018	0.0400	928.94
β_4 (Skin Thickness)	0.2364	0.2864	0.0048	0.8521	675.96
β_5 (Insulin)	0.1684	0.1641	0.0080	0.5275	447.91
β_6 (BMI)	0.0473	.	0.0047	0.1828	490.99
β_7 (Age)	0.0161	.	0.0021	0.0579	1000.0
σ^2	0.1111	0.1414	0.0002	.	1160.29

Table 2: Model Output

already imposes sparsity on the model, and significant shrinkage is likely unnecessary for a model with a small number of predictors.

3 Results

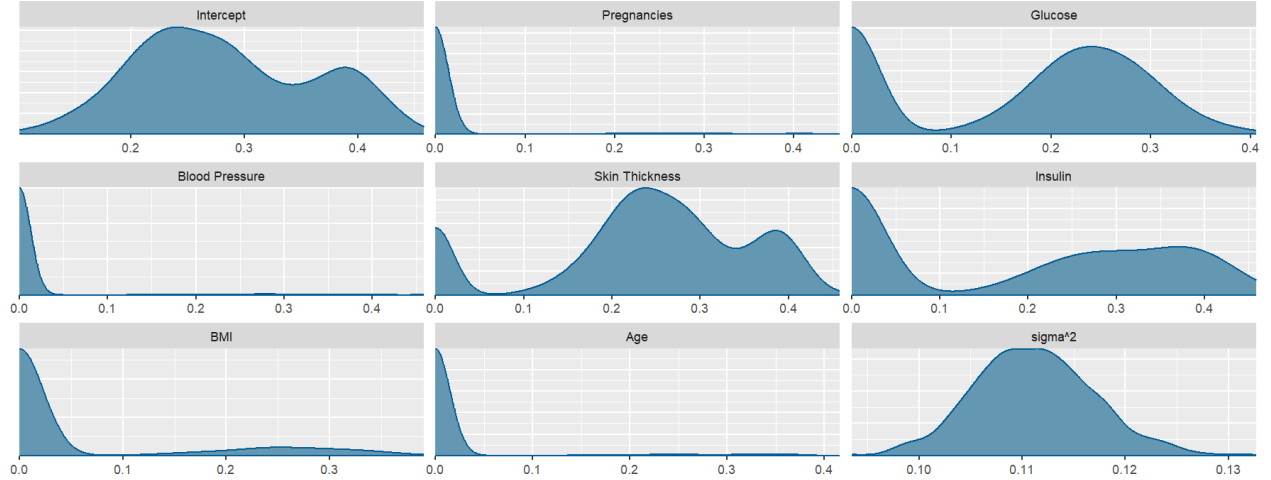


Figure 4: Posterior Density Plots

Based on the posterior distributions of each parameter obtained from the Gibbs sampler (see Figure 4) we see that each coefficient has a Gaussian or Gaussian mixture distribution, except with a spike at zero as a result of the spike and slab prior. The posterior distribution for σ^2 has a bell curve shape that suggests the variance in a given patient's DPF due to factors not considered by the model is reasonably between 0.100 and 0.124 (based on a 95% credible interval).

Notably, the Intercept and Skin Thickness distributions appear to be bimodal, which may suggest that the samples belong to 2 differing groups with a hierarchy of distributions in y and β , however, there is not enough information in the dataset to make conclusions about what the cause for the hierarchy is. It may be the case that there are two distinct genetic groupings for women of Pima Indian descent with different average DPF values, and depending on what group a patient belongs to, their triceps skin fold thickness has a different relationship with genetic diabetic risk. This conclusion is a conjecture and may need further study.

The size of the spike at zero in each coefficient distribution is reflective of the posterior inclusion probabilities (see Table 2). The inclusion probabilities suggest that the glucose levels, insulin levels, and skin thickness are the most significant predictors ($PIP > 0.5$) for one’s diabetes pedigree function, while number of pregnancies, blood pressure, BMI, and age are not as significant to the overall model. Using the resulting MPM coefficients, a separate model was fit with the same g-prior (but with z fixed) from which we can estimate each parameter using their posterior mean to obtain the predictive model:

$$\begin{aligned} \text{DPF} &= 0.267 + .0102(\text{Glucose}) + 0.286(\text{Skin Thickness}) + 0.164(\text{Insulin}) + \epsilon \\ \epsilon &\sim \mathcal{N}(0, 0.141) \end{aligned}$$

which can be interpreted as follows: holding constant for other variables, a unitary increase in plasma glucose levels corresponds with a increase in DPF of 0.102, and likewise a millimeter increase in triceps skin fold thickness corresponds with an increase in DPF score of 0.286, and a $\mu\text{U}/\text{mm}$ increase in insulin levels corresponds with an increase in DPF score of 0.164.

To judge the predictive performance of the BMA and MPM, the validation set was used to create predicted DFP values $\hat{y}_{BMA}^{test}$ and $\hat{y}_{MPM}^{test}$, and these were compared with the observed values in the validation set y^{test} to obtain prediction error metrics (see Table 3). The BMA obtains a smaller mean square error ($MSE = \|y^{test} - \hat{y}_{BMA}^{test}\|_2^2/n_{test}$) and mean absolute error ($MAE = \|y^{test} - \hat{y}_{BMA}^{test}\|_1/n_{test}$), which suggests it is optimal for DPF prediction compared to the MPM. However, the MPM may still be a desirable model for its simpler interpretation and reliance on fewer predictors. In conclusion, we see in the case that genetic history is not available to formally

Model	MSE	MAE
BMA	0.0882	0.3956
MPM	0.0988	0.2417

Table 3: Prediction Analysis

estimate the DPF for a woman of Pima Indian descent, a reasonable estimate can be create from a relatively simple model that uses diagnostic measurements. In general, the glucose levels, insulin levels, and skin thickness of a patient are the most useful predictors for estimating a patients DPF, which suggests they may be related to phenotypes characterizing the genetic risk of diabetes in a patient. In contrast, other factors such as blood pressure or BMI may not provide as much information about one’s genetic risk.

Algorithm 1 Gibbs sampler for β, z, σ^2

```
initialize  $z_j^{(0)} \sim \text{Bern}(1/2), j = 1, \dots, p$ 
for  $s = 1, \dots$  do
  for  $j = 1, \dots, p$  do
    compute  $\pi = \frac{o_j}{1+o_j}$ 
    sample  $z_j^{(s)} \sim \text{Bern}(\pi)$ 
  end for
  let  $z^{(s)} = [z_1^{(s)} \dots z_p^{(s)}]^\top$ 
  update  $X_z, \hat{b}_z, SSR_g^z$ 
  sample  $\sigma^{2(s)} \sim \text{IG}(\frac{\nu_0+n}{2}, \frac{\nu_0\sigma_0^2+SSR_g^z}{2})$ 
  sample  $\beta^{(s)} \sim \mathcal{N}(\frac{g}{g+1}\hat{b}_z, \frac{g}{g+1}\sigma^{2(s)}[X_z^\top X_z]^{-1})$ 
end for
```
